COMPUTATIONAL STATISTICS

SUMMARY

WEEK SIX

The Gibbs sampling idea can be thought of as an algorithm that generates samples from a multivariate probability distribution when drawing directly from the joint distribution is challenging but sampling from its conditional distributions is more feasible. Thus, a Gibbs sampler is a Markov chain whose transition kernel is built by composing conditional updates. It can be viewed as a sequence of Metropolis-Hastings (MH) updates with acceptance probability of 1. We did some computations of the Gibbs sampler in R. We discovered that a downside of the Gibbs sampler is that it can be inefficient when the parameters are strongly correlated. Again, we learnt that a valid MCMC algorithm can have the correct target distribution as its stationary distribution and still be computationally inefficient.

Hamiltonian Monte Carlo (HMC) differs from the other MCMC methods in that it uses the gradient of the log target density to guide its proposals. The main geometric idea behind the HMC is that the dynamics can travel a long distance through parameter space without having to cross into much lower joint density. Because these trajectories can travel far while maintaining a high acceptance probability, HMC often produces less correlated samples, particularly in high-dimensional, smooth distributions. However, each iteration costs more because it requires several gradient evaluations. Standard HMC also requires a differentiable target over continuous variables, so it does not directly update discrete variables. Later in the lecture, we looked at how NUTS works. We established that, under HMC, a trajectory can begin to turn back on itself. NUTS (No-U-Turn Sampler) builds a trajectory adaptively and uses a geometric U-turn criterion to decide when continued expansion is no longer useful and it also addresses the problem of tuning with the HMC. Lastly, we learnt that Stan is not a substitute for the model, but it makes sophisticated MCMC accessible. It is incumbent on the statistician to design the model.

The second lecture covered variational inference (VI). Variational inference takes a different approach from Bayesian statistics and MCMC. It approximates probability distribution through optimization and not sampling. It serves the same purpose as traditional MCMC algorithms. The idea behind VI is to first posit a family of distributions and then to find the member of that family which is close to the target, where closeness is measured using the Kullback-Leibler divergence.

The Kullback-Leibler divergence involves two steps called Expectation-Maximization (EM) algorithm. In the E-step we learn about latient variables and in the M-step our goal is to maximize the parameter from our knowledge of the latent variables. The advantage of VI is that it can be much faster, but it introduces a new source of error called approximation error derived from the variational family and optimization. ELBO, Jensen's inequality derivation, Mean-field variational inference and coordinate ascent variational inference are some VI approaches we learned in the lecture.

In conclusion, the concepts covered in the two lectures built very much on what we had already seen in the course which enhanced understanding. My only confusion was with the difference between CAVI and the Gibbs sampler.